

3.5 PRACTICE SOLUTIONS

1. Prove each identity.

a) $\frac{\sec(\theta) - 1}{1 - \cos(\theta)} = \sec(\theta)$

$$\begin{aligned} LS &= \frac{\frac{1}{\cos \theta} - \frac{\cos \theta}{\cos \theta}}{1 - \cos \theta} \\ &= \frac{\frac{1 - \cos \theta}{\cos \theta}}{1 - \cos \theta} \cdot \frac{1}{1 - \cos \theta} \\ &= \frac{1}{\cos \theta} \\ &= \sec \theta \\ &= RS \end{aligned}$$

b) $\tan(x) + \cot(x) = \frac{2}{\sin(2x)}$

$$\begin{aligned} LS &= \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} \\ &= \frac{\sin^2 x + \cos^2 x}{\cos x \sin x} \\ &= \frac{1}{\frac{1}{2} \sin(2x)} \\ &= \frac{2}{\sin(2x)} \\ &= RS \end{aligned}$$

c) $\frac{1 + \tan^2(x)}{1 + \cot^2(x)} = \frac{1 - \cos^2(x)}{\cos^2(x)}$

$$\begin{aligned} LS &= \frac{1 + \frac{\sin^2 x}{\cos^2 x}}{1 + \frac{\cos^2 x}{\sin^2 x}} \\ &= \frac{\frac{\cos^2 x + \sin^2 x}{\cos^2 x}}{\frac{\sin^2 x + \cos^2 x}{\sin^2 x}} \\ &= \frac{1}{\cos^2 x} \cdot \frac{\sin^2 x}{1} \\ &= \frac{\sin^2 x}{\cos^2 x} \\ &= \frac{1 - \cos^2 x}{\cos^2 x} \\ &= RS \end{aligned}$$

d) $\frac{1}{1 + \sec(\theta)} + \frac{1}{1 - \sec(\theta)} = -2\cot^2(\theta)$

$$\begin{aligned} LS &= \frac{1 - \sec \theta + 1 + \sec \theta}{(1 + \sec \theta)(1 - \sec \theta)} \\ &= \frac{2}{1 - \sec^2 \theta} \\ &= \frac{2}{1 - \frac{1}{\cos^2 \theta}} \\ &= \frac{2}{\frac{\cos^2 \theta - 1}{\cos^2 \theta}} \\ &= \frac{2 \cos^2 \theta}{\cos^2 \theta - 1} \\ &= \frac{2 \cos^2 \theta}{-\sin^2 \theta} \\ &= -2\cot^2 \theta \\ &= RS \end{aligned}$$

e) $\frac{1 + \sec(x)}{\tan(x) + \sin(x)} = \csc(x)$

$$\begin{aligned} LS &= \frac{1 + \frac{1}{\cos x}}{\frac{\sin x}{\cos x} + \sin x} \\ &= \frac{\frac{\cos x + 1}{\cos x}}{\frac{\sin x + \sin x \cdot \cos x}{\cos x}} \\ &= \frac{\cos x + 1}{\cancel{\cos x}} \cdot \frac{\cancel{\cos x}}{\sin x (1 + \cos x)} \\ &= \frac{1}{\sin x} \\ &= \csc x \\ &= RS \end{aligned}$$

f) $\cos(a+b)\cos(a-b) = \cos^2(a) - \sin^2(b)$

$$\begin{aligned} LS &= [\cos(a)\cos(b) - \sin(a)\sin(b)][\cos(a)\cos(b) + \sin(a)\sin(b)] \\ &= \cos^2(a)\cos^2(b) - \sin^2(a)\sin^2(b) \\ &= \cos^2(a)[1 - \sin^2(b)] - [1 - \cos^2(a)]\sin^2(b) \\ &= \cos^2(a) - \cancel{\cos^2(a)\sin^2(b)} - \sin^2(b) + \cancel{\cos^2(a)\sin^2(b)} \\ &= \cos^2(a) - \sin^2(b) \\ &= RS \end{aligned}$$

$$g) \frac{\cos(x) - \sin(y)}{\cos(y) - \sin(x)} = \frac{\cos(y) + \sin(x)}{\cos(x) + \sin(y)}$$

$$\begin{aligned} LS &= \frac{\cos(x) - \sin(y)}{\cos(y) - \sin(x)} \cdot \frac{\cos(x) + \sin(y)}{\cos(x) + \sin(y)} \\ &= \frac{\cos^2(x) - \sin^2(y)}{[\cos(y) - \sin(x)][\cos(x) + \sin(y)]} \\ &= \frac{1 - \sin^2(x) - (1 - \cos^2(y))}{[\cos(y) - \sin(x)][\cos(x) + \sin(y)]} \\ &= \frac{1 - \sin^2(x) - 1 + \cos^2(y)}{[\cos(y) - \sin(x)][\cos(x) + \sin(y)]} \\ &= \frac{\cos^2(y) - \sin^2(x)}{[\cos(y) - \sin(x)][\cos(x) + \sin(y)]} \\ &= \frac{\cancel{[\cos(y) - \sin(x)]} [\cos(y) + \sin(x)]}{\cancel{[\cos(y) - \sin(x)]} [\cos(x) + \sin(y)]} \\ &= \frac{\cos(y) + \sin(x)}{\cos(x) + \sin(y)} \\ &= RS \end{aligned}$$

$$h) \frac{1 + \tan(A)}{\sin(A)} - \sec(A) = \csc(A)$$

$$\begin{aligned} LS &= \frac{1 + \frac{\sin A}{\cos A}}{\sin A} - \frac{1}{\cos A} \\ &= \frac{\frac{\cos A + \sin A}{\cos A}}{\sin A} - \frac{1}{\cos A} \\ &= \frac{\cos A + \sin A}{\sin A \cos A} - \frac{1}{\cos A} \\ &= \frac{\cos A + \sin A - \sin A}{\sin A \cos A} \\ &= \frac{\cos A}{\sin A \cos A} \\ &= \frac{1}{\sin A} \\ &= \csc A \\ &= RS \end{aligned}$$

$$i) \frac{\sin(t) - \cos(t)}{\cos(t)} + \frac{\sin(t) + \cos(t)}{\sin(t)} = \sec(t) \csc(t)$$

$$\begin{aligned} LS &= \frac{\sin(t)[\sin(t) - \cos(t)] + \cos(t)[\sin(t) + \cos(t)]}{\cos(t) \sin(t)} \\ &= \frac{\sin^2(t) - \cancel{\sin(t)\cos(t)} + \cancel{\cos(t)\sin(t)} + \cos^2(t)}{\cos(t) \sin(t)} \\ &= \frac{\sin^2(t) + \cos^2(t)}{\cos(t) \sin(t)} \\ &= \frac{1}{\cos(t) \sin(t)} \\ &= \sec(t) \csc(t) \\ &= RS \end{aligned}$$

$$j) \tan(A) + \cot(A) = \sec^2(A) \cot(A)$$

$$\begin{aligned} LS &= \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} \\ &= \frac{\sin^2 A + \cos^2 A}{\cos A \sin A} \\ &= \frac{1}{\cos A \sin A} \cdot \frac{\cos A}{\cos A} \\ &= \frac{\cos A}{\cos^2 A \sin A} \\ &= \left(\frac{1}{\cos^2 A} \right) \left(\frac{\cos A}{\sin A} \right) \\ &= \sec^2(A) \cot(A) \\ &= RS \end{aligned}$$

$$k) \frac{4 - \sin^2(2x)}{4\cos^4(x)} = \tan^4(x) + \tan^2(x) + 1$$

$$\begin{aligned} LS &= \frac{4 - [\sin(2x)\cos(x)]^2}{4\cos^4(x)} \\ &= \frac{4 - 4\sin^2(x)\cos^2(x)}{4\cos^4(x)} \\ &= \frac{4[1 - \sin^2(x)\cos^2(x)]}{4\cos^4(x)} \\ &= \frac{1}{\cos^4(x)} - \frac{\sin^2(x)\cos^2(x)}{\cos^4(x)} \\ &= \frac{(\cos^2(x) + \sin^2(x))^2}{\cos^4(x)} - \frac{\sin^2(x)\cos^2(x)}{\cos^4(x)} \\ &= \frac{\cos^4(x) + 2\sin^2(x)\cos^2(x) + \sin^4(x)}{\cos^4(x)} - \frac{\sin^2(x)\cos^2(x)}{\cos^4(x)} \\ &= \frac{\cos^4(x)}{\cos^4(x)} + \frac{2\sin^2(x)\cos^2(x)}{\cos^4(x)} + \frac{\sin^4(x)}{\cos^4(x)} - \frac{\sin^2(x)\cos^2(x)}{\cos^4(x)} \\ &= 1 + \frac{\sin^2(x)\cos^2(x)}{\cos^4(x)} + \tan^4(x) \\ &= \tan^4(x) + \frac{\sin^2(x)}{\cos^2(x)} + 1 \\ &= \tan^4(x) + \tan^2(x) + 1 \\ &= RS \end{aligned}$$

$$m) \frac{\sin(x-y)}{\sin(x)\sin(y)} = \cot(y) - \cot(x)$$

$$\begin{aligned} LS &= \frac{\sin(x)\cos(y) - \cos(x)\sin(y)}{\sin(x)\sin(y)} \\ &= \frac{\sin(x)\cos(y)}{\sin(x)\sin(y)} - \frac{\cos(x)\sin(y)}{\sin(x)\sin(y)} \\ &= \frac{\cos(y)}{\sin(y)} - \frac{\cos(x)}{\sin(x)} \\ &= \cot(y) - \cot(x) \\ &= RS \end{aligned}$$

$$l) 1 - \sin(x)\cos(x) = \frac{\sin^2(x)}{1 + \cot(x)} + \frac{\cos^2(x)}{1 + \tan(x)}$$

$$\begin{aligned} RS &= \frac{\sin^2(x)}{1 + \frac{\cos(x)}{\sin(x)}} + \frac{\cos^2(x)}{1 + \frac{\sin(x)}{\cos(x)}} \\ &= \frac{\sin^2(x)}{\frac{\sin(x) + \cos(x)}{\sin(x)}} + \frac{\cos^2(x)}{\frac{\cos(x) + \sin(x)}{\cos(x)}} \\ &= \sin^2(x) \cdot \frac{\sin(x)}{\sin(x) + \cos(x)} + \cos^2(x) \cdot \frac{\cos(x)}{\cos(x) + \sin(x)} \\ &= \frac{\sin^3(x)}{\sin(x) + \cos(x)} + \frac{\cos^3(x)}{\sin(x) + \cos(x)} \\ &= \frac{\sin^3(x) + \cos^3(x)}{\sin(x) + \cos(x)} \quad \text{a}^3 + b^3 = (a+b)(a^2 - ab + b^2) \\ &= \frac{[\sin(x) + \cos(x)][\sin^2(x) - \sin(x)\cos(x) + \cos^2(x)]}{\sin(x) + \cos(x)} \\ &= 1 - \sin(x)\cos(x) \\ &= RS \end{aligned}$$

$$n) \frac{\sin(5x)}{\sin(x)} - \frac{\cos(5x)}{\cos(x)} = 4 - 8\sin^2(x)$$

$$\begin{aligned} LS &= \frac{\sin(5x)\cos(x) - \cos(5x)\sin(x)}{\sin(x)\cos(x)} \\ &= \frac{\sin(5x - x)}{\sin(x)\cos(x)} \\ &= \frac{\sin(4x)}{\sin(x)\cos(x)} \\ &= \frac{2\sin(2x)\cos(2x)}{\sin(x)\cos(x)} \\ &= \frac{2[\sin(x)\cos(x)][1 - 2\sin^2(x)]}{\sin(x)\cos(x)} \\ &= 4[1 - 2\sin^2(x)] \\ &= 4 - 8\sin^2(x) \\ &= RS \end{aligned}$$